

Exercises

4.6-1

Show that $\sum_{j=0}^{\lfloor \log_b n \rfloor} (\log_b n - j)^k = \Omega(\log_b^{k+1} n)$.

★ 4.6-2

Show that case 3 of the master theorem is overstated (which is also why case 3 of Lemma 4.3 does not require that $f(n) = \Omega(n^{\log_b a + \epsilon})$) in the sense that the regularity condition $af(n/b) \leq cf(n)$ for some constant $c < 1$ implies that there exists a constant $\epsilon > 0$ such that $f(n) = \Omega(n^{\log_b a + \epsilon})$.

★ 4.6-3

For $f(n) = \Theta(n^{\log_b a} / \lg n)$, prove that the summation in equation (4.19) has solution $g(n) = \Theta(n^{\log_b a} \lg \lg n)$. Conclude that a master recurrence $T(n)$ using $f(n)$ as its driving function has solution $T(n) = \Theta(n^{\log_b a} \lg \lg n)$.

★ 4.7 Akra-Bazzi recurrences

This section provides an overview of two advanced topics related to divide-and-conquer recurrences. The first deals with technicalities arising from the use of floors and ceilings, and the second discusses the Akra-Bazzi method, which involves a little calculus, for solving complicated divide-and-conquer recurrences.

In particular, we'll look at the class of algorithmic divide-and-conquer recurrences originally studied by M. Akra and L. Bazzi [13]. These *Akra-Bazzi* recurrences take the form

$$T(n) = f(n) + \sum_{i=1}^k a_i T(n/b_i), \quad (4.22)$$

where k is a positive integer; all the constants $a_1, a_2, \dots, a_k \in \mathbb{R}$ are strictly positive; all the constants $b_1, b_2, \dots, b_k \in \mathbb{R}$ are strictly greater than 1; and the driving function $f(n)$ is defined on sufficiently large nonnegative reals and is itself nonnegative.

Akra-Bazzi recurrences generalize the class of recurrences addressed by the master theorem. Whereas master recurrences

characterize the running times of divide-and-conquer algorithms that break a problem into equal-sized subproblems (modulo floors and ceilings), Akra-Bazzi recurrences can describe the running time of divide-and-conquer algorithms that break a problem into different-sized subproblems. The master theorem, however, allows you to ignore floors and ceilings, but the Akra-Bazzi method for solving Akra-Bazzi recurrences needs an additional requirement to deal with floors and ceilings.

But before diving into the Akra-Bazzi method itself, let's understand the limitations involved in ignoring floors and ceilings in Akra-Bazzi recurrences. As you're aware, algorithms generally deal with integer-sized inputs. The mathematics for recurrences is often easier with real numbers, however, than with integers, where we must cope with floors and ceilings to ensure that terms are well defined. The difference may not seem to be much—especially because that's often the truth with recurrences—but to be mathematically correct, we must be careful with our assumptions. Since our end goal is to understand algorithms and not the vagaries of mathematical corner cases, we'd like to be casual yet rigorous. How can we treat floors and ceilings casually while still ensuring rigor?

From a mathematical point of view, the difficulty in dealing with floors and ceilings is that some driving functions can be really, really weird. So it's not okay in general to ignore floors and ceilings in Akra-Bazzi recurrences. Fortunately, most of the driving functions we encounter in the study of algorithms behave nicely, and floors and ceilings don't make a difference.

The polynomial-growth condition

If the driving function $f(n)$ in equation (4.22) is well behaved in the following sense, it's okay to drop floors and ceilings.

A function $f(n)$ defined on all sufficiently large positive reals satisfies the **polynomial-growth condition** if there exists a constant $\hat{n} > 0$ such that the following holds: for every constant

$\phi \geq 1$, there exists a constant $d > 1$ (depending on ϕ) such that $f(n)/d \leq f(\psi n) \leq df(n)$ for all $1 \leq \psi \leq \phi$ and $n \geq \hat{n}$.

This definition may be one of the hardest in this textbook to get your head around. To a first order, it says that $f(n)$ satisfies the property that $f(\Theta(n)) = \Theta(f(n))$, although the polynomial-growth condition is actually somewhat stronger (see Exercise 4.7-4). The definition also implies that $f(n)$ is asymptotically positive (see Exercise 4.7-3).

Examples of functions that satisfy the polynomial-growth condition include any function of the form $f(n) = \Theta(n^\alpha \lg^\beta n \lg^\gamma n)$, where α , β , and γ are constants. Most of the polynomially bounded functions used in this book satisfy the condition. Exponentials and superexponentials do not (see Exercise 4.7-2, for example), and there also exist polynomially bounded functions that do not.

Floors and ceilings in “nice” recurrences

When the driving function in an Akra-Bazzi recurrence satisfies the polynomial-growth condition, floors and ceilings don’t change the asymptotic behavior of the solution. The following theorem, which is presented without proof, formalizes this notion.

Theorem 4.5

Let $T(n)$ be a function defined on the nonnegative reals that satisfies recurrence (4.22), where $f(n)$ satisfies the polynomial-growth condition. Let $T'(n)$ be another function defined on the natural numbers also satisfying recurrence (4.22), except that each $T(n/b_i)$ is replaced either with $T(\lceil n/b_i \rceil)$ or with $T(\lfloor n/b_i \rfloor)$. Then we have $T'(n) = \Theta(T(n))$.

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Floors and ceilings represent a minor perturbation to the arguments in the recursion. By inequality (3.2) on page 64, they perturb an argument by at most 1. But much larger perturbations are tolerable. As long as the driving function $f(n)$ in recurrence (4.22) satisfies the polynomial-growth condition, it turns out that replacing any term T

(n/b_i) with $T(n/b_i + h_i(n))$, where $|h_i(n)| = O(n/\lg^{1+\epsilon} n)$ for some constant $\epsilon > 0$ and sufficiently large n , leaves the asymptotic solution unaffected. Thus, the divide step in a divide-and-conquer algorithm can be moderately coarse without affecting the solution to its running-time recurrence.

The Akra-Bazzi method

The Akra-Bazzi method, not surprisingly, was developed to solve Akra-Bazzi recurrences (4.22), which by dint of Theorem 4.5, applies in the presence of floors and ceilings or even larger perturbations, as just discussed. The method involves first determining the unique real number p such that $\sum_{i=1}^k a_i/b_i^p = 1$. Such a p always exists, because when $p \rightarrow -\infty$, the sum goes to ∞ ; it decreases as p increases; and when $p \rightarrow \infty$, it goes to 0. The Akra-Bazzi method then gives the solution to the recurrence as

$$T(n) = \Theta\left(n^p \left(1 + \int_1^n \frac{f(x)}{x^{p+1}} dx\right)\right). \quad (4.23)$$

As an example, consider the recurrence

$$T(n) = T(n/5) + T(7n/10) + n. \quad (4.24)$$

We'll see the similar recurrence (9.1) on page 240 when we study an algorithm for selecting the i th smallest element from a set of n numbers. This recurrence has the form of equation (4.22), where $a_1 = a_2 = 1$, $b_1 = 5$, $b_2 = 10/7$, and $f(n) = n$. To solve it, the Akra-Bazzi method says that we should determine the unique p satisfying

$$\left(\frac{1}{5}\right)^p + \left(\frac{7}{10}\right)^p = 1.$$

Solving for p is kind of messy—it turns out that $p = 0.83978 \dots$ —but we can solve the recurrence without actually knowing the exact value for p . Observe that $(1/5)^0 + (7/10)^0 = 2$ and $(1/5)^1 + (7/10)^1 = 9/10$, and thus p lies in the range $0 < p < 1$. That turns out to be sufficient for the Akra-Bazzi method to give us the solution. We'll use the fact from

calculus that if $k \neq -1$, then $\int x^k dx = x^{k+1}/(k+1)$, which we'll apply with $k = -p \neq -1$. The Akra-Bazzi solution (4.23) gives us

$$\begin{aligned}
 T(n) &= \Theta\left(n^p \left(1 + \int_1^n \frac{f(x)}{x^{p+1}} dx\right)\right) \\
 &= \Theta\left(n^p \left(1 + \int_1^n x^{-p} dx\right)\right) \\
 &= \Theta\left(n^p \left(1 + \left[\frac{x^{1-p}}{1-p}\right]_1^n\right)\right) \\
 &= \Theta\left(n^p \left(1 + \left(\frac{n^{1-p}}{1-p} - \frac{1}{1-p}\right)\right)\right) \\
 &= \Theta\left(n^p \cdot \Theta(n^{1-p})\right) && \text{(because } 1-p \text{ is a positive constant)} \\
 &= \Theta(n) && \text{(by Problem 3-5(d)) .}
 \end{aligned}$$

Although the Akra-Bazzi method is more general than the master theorem, it requires calculus and sometimes a bit more reasoning. You also must ensure that your driving function satisfies the polynomial-growth condition if you want to ignore floors and ceilings, although that's rarely a problem. When it applies, the master method is much simpler to use, but only when subproblem sizes are more or less equal. They are both good tools for your algorithmic toolkit.

Exercises

★ 4.7-1

Consider an Akra-Bazzi recurrence $T(n)$ on the reals as given in recurrence (4.22), and define $T'(n)$ as

$$T'(n) = cf(n) + \sum_{i=1}^k a_i T'(n/b_i),$$

where $c > 0$ is constant. Prove that whatever the implicit initial conditions for $T(n)$ might be, there exist initial conditions for $T'(n)$ such that $T'(n) = cT(n)$ for all $n > 0$. Conclude that we can drop the asymptotics on a driving function in any Akra-Bazzi recurrence without affecting its asymptotic solution.

4.7-2

Show that $f(n) = n^2$ satisfies the polynomial-growth condition but that $f(n) = 2^n$ does not.

4.7-3

Let $f(n)$ be a function that satisfies the polynomial-growth condition. Prove that $f(n)$ is asymptotically positive, that is, there exists a constant $n_0 \geq 0$ such that $f(n) \geq 0$ for all $n \geq n_0$.

★ 4.7-4

Give an example of a function $f(n)$ that does not satisfy the polynomial-growth condition but for which $f(\Theta(n)) = \Theta(f(n))$.

4.7-5

Use the Akra-Bazzi method to solve the following recurrences.

a. $T(n) = T(n/2) + T(n/3) + T(n/6) + n \lg n.$

b. $T(n) = 3T(n/3) + 8T(n/4) + n^2/\lg n.$

c. $T(n) = (2/3)T(n/3) + (1/3)T(2n/3) + \lg n.$

d. $T(n) = (1/3)T(n/3) + 1/n.$

e. $T(n) = 3T(n/3) + 3T(2n/3) + n^2.$

★ 4.7-6

Use the Akra-Bazzi method to prove the continuous master theorem.

Problems
4-1 Recurrence examples

Give asymptotically tight upper and lower bounds for $T(n)$ in each of the following algorithmic recurrences. Justify your answers.

a. $T(n) = 2T(n/2) + n^3.$

b. $T(n) = T(8n/11) + n.$